

Basic Concepts

A serious discussion of aeronautics requires a basic set of concepts and terminology. Frequently, in the process of learning, the language becomes an unrecognized barrier to the “uninitiated.” By starting with some basic terminology and concepts we hope to alleviate some of this.

Airplane Nomenclature

Some readers may be familiar with the language of airplanes and others not. We encourage all to read the following sections of this chapter to ensure that subsequent chapters are easily understood. Those familiar with the major parts of the airplane, the operation of the control surfaces, and the basic operation of an airplane can skip to the section on kinetic energy.

There are over 600,000 licensed pilots in the United States.

The Airplane

Figure 1.1 shows the main components of a high-winged airplane. The *airframe* consists of the *fuselage*, which is the main component of the airplane, the *wings*, and the *empennage*. The empennage (sometimes called the *tail feathers*) is the tail assembly consisting of the *horizontal stabilizer*, the *elevators*, the *vertical stabilizer*, and the

rudder. The elevators are used to adjust, or control, the pitch (nose up/down attitude) of the airplane. The elevators are connected to the control wheel or stick of the airplane and are moved by the forward and backward motion of the control. On some airplanes the entire horizontal stabilizer is the elevator, as shown in Figure 1.2. This is called a *stabilator*. The rudder is used to make small directional changes and in turns. Two pedals on the floor operate the rudder, used to provide directional control.

Most airplanes have small hinged sections on the trailing edge of the elevators and sometimes on the rudder called *trim tabs* as shown in Figure 1.2. These tabs move in the opposite direction to the control

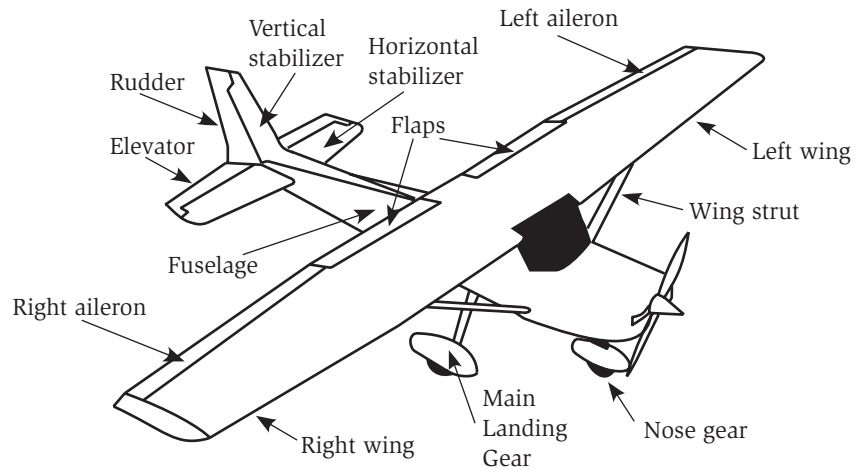


Fig. 1.1. Main components of an airplane.

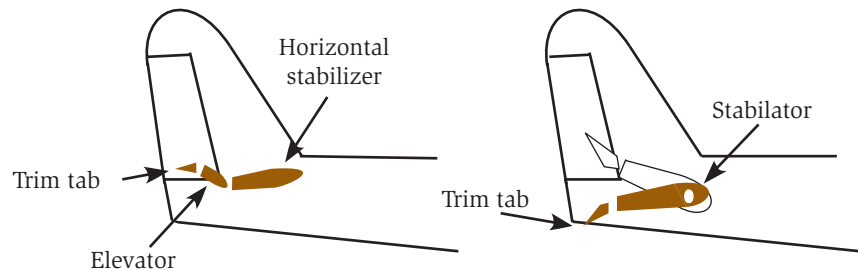


Fig. 1.2. The empennage.

surface. The purpose of the trim tabs is to reduce the necessary force on the control wheel, called a *yoke*, for the pilot to maintain a desired flight attitude.

Most modern airplanes have single wings mounted either above or below the fuselage. Most but not all high-winged airplanes have wings that are supported by *struts*. Struts allow for a lighter wing but at the expense of more drag (resistance to motion through the air).

The movable surfaces on the outer trailing edge of the wings are the *ailerons*, which are used for roll control (rotation around the axis of the fuselage). They are operated by the rotation of the control wheel or by the left-right movement of the stick. The ailerons are coupled so that when one swings up the other swings down. Control surfaces are discussed in detail below.

A tail dragger is also known as conventional gear because before WWII nose wheels were rare.

The hinged portions on the inboard part of the trailing edge of the wings are the *flaps*. These are used to produce greater lift at low speeds and to provide increased drag on landing. This increased drag helps to reduce the speed of the airplane and to steepen the landing approach angle. Flaps are discussed in detail in the chapter on wings.

Small airplanes have two configurations of landing gear. *Tricycle landing gear* has the *main landing gear* just behind the center of balance of the airplane and a steerable *nose gear* up forward. The *tail dragger* has the main landing gear forward of the center of balance and a small steerable wheel at the tail. The nose gear and the tail wheel are steered with the rudder pedals.

Airfoils and Wings

An *airfoil* is a shape designed to produce lift. As shown in Figure 1.3, an airfoil is the shape seen in a slice of a wing. Besides the wing, propellers and the tail surfaces are also airfoils. Even aeronautical engineers (in discussions) sometimes mistakenly use the terms *wing* and *airfoil* interchangeably. But an airfoil is just the shape seen in a slice of the wing and not a wing itself. For some wings, slices taken at different places along its length will reveal different airfoils.

An airfoil, as shown in Figure 1.3, has a *leading edge* and a *trailing edge*. As detailed in Figure 1.4, a *chord* and a *camber* also characterize an airfoil. The chord is an imaginary straight line connecting the

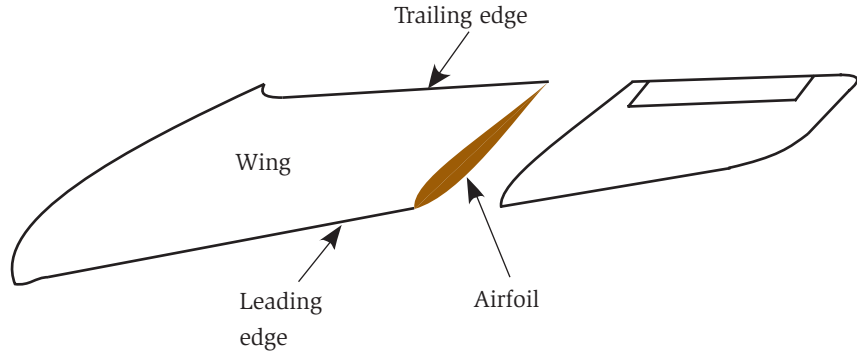


Fig. I.3. The wing and airfoil.

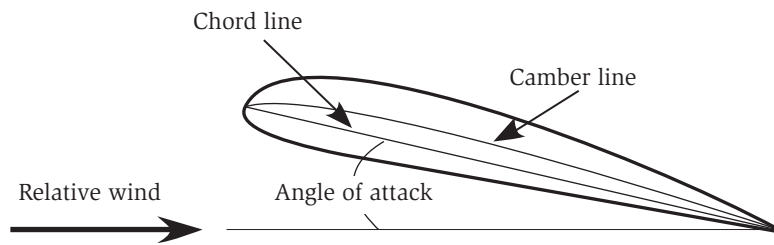


Fig. I.4. Airfoil nomenclature.

leading edge with the trailing edge. The chord is used for determining the geometric *angle of attack* (discussed below) and for determining the area of a wing.

The *mean camber line* is the line an equal distance from the upper and lower surfaces of the wing. The camber is the curvature of the mean camber line. A wing that has an airfoil with a great deal of curvature in its mean camber line is said to be a *highly cambered* wing. A symmetric airfoil has no camber.

An airfoil with lift also has an angle of attack, as shown in the figure. The *relative wind* is the direction of the wind at some distance from the wing. It is parallel to and opposite to the direction of motion of the wing. The velocity of the relative wind is equal to the speed of the wing. In aeronautics the *geometric angle of attack* is defined as the angle between the *mean chord* of the airfoil and the direction of the relative wind.

A useful measure of a wing is its *aspect ratio*. The aspect ratio is defined as the wing's *span* divided by the average or *mean chord* length. The span is the length of the wing measured from wingtip to wingtip. The mean chord length is the average chord length along the wing. The area of the wing is just the span times the mean chord length. Most wings on small general-aviation airplanes have aspect ratios of about 6 to 8. This means that the wing is 6 to 8 times longer than its average width.

The Boeing B-314 Flying Boats were so large that there was a catwalk in the wings so the mechanics could service the engines in flight.

Axes of Control

An airplane moves in three dimensions called *roll*, *pitch*, and *yaw*, illustrated in Figure 1.5. Roll is rotation about the *longitudinal axis* that goes down the center of the fuselage. The ailerons control rotation about the roll axis. Pitch is rotation about the lateral axis of rotation, which is an axis parallel to the long dimension of the wings. The elevators control the pitch of the airplane. By controlling the pitch of the airplane, the elevators also control the angle of attack of the wing. To increase the angle of attack, the entire airplane is rotated up. As we will see, this control or the angle of attack is key in the adjustment of

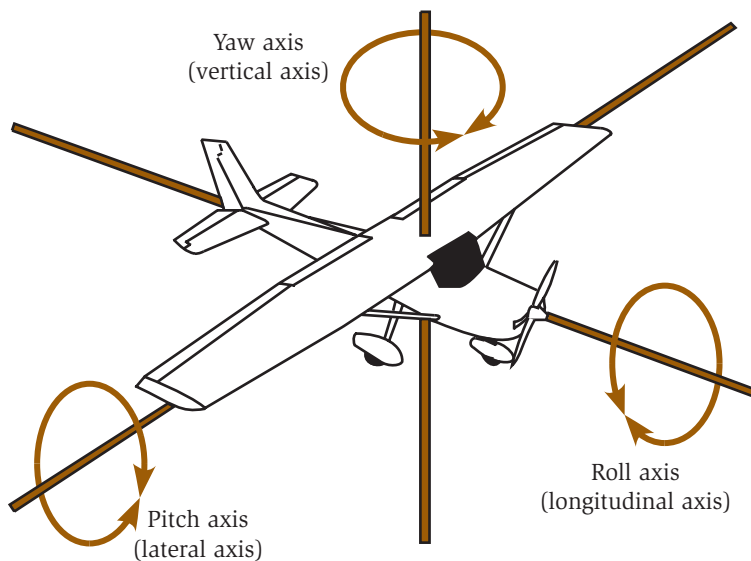


Fig. 1.5. Axis of rotation of an airplane.

the lift of the wings. Finally, yaw, which is controlled by the rudder, is rotation about the vertical axis, which is a line that goes vertically through the center of the wing. It is important to note that all three axes go through the *center of gravity* (often abbreviated c.g.) of the airplane. The center of gravity is the balance point of the airplane. Or, equivalently, all of the weight of the airplane can be considered to be at that one point.

The Turn

The first flight attendants (stewardesses) were required to be registered nurses.

One common misconception by those who are not pilots is that, as with a boat, the rudder is the control used for making a turn. Although very small direction changes can be made with the rudder, the ailerons are used in making turns. The airplane is rolled to an angle in the direction of the desired turn. The lift developed is perpendicular to the top of the wing. In straight-and-level flight, this is straight up. As shown in Figure 1.6, when the airplane rolls to some angle, the direction of lift is now at an angle with part of the lift force used for turning and part used to support the weight of the airplane. In a turn the rudder is only used to make small corrections and *coordinate* the turn.

As shown in the figure, the pilot feels a force equal to the lift but in the opposite direction. Occasionally in the following chapters we refer to the $2g$ turn. A $2g$ turn is a turn where the force felt by the pilot is twice the force of gravity ($2g$) and the force, or *load*, on the wing has been doubled. In aeronautical terms the *load factor*, which is the

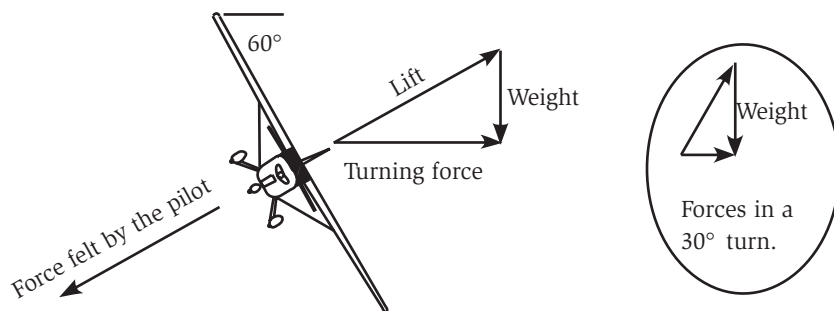


Fig. 1.6. Forces on an airplane in a turn.

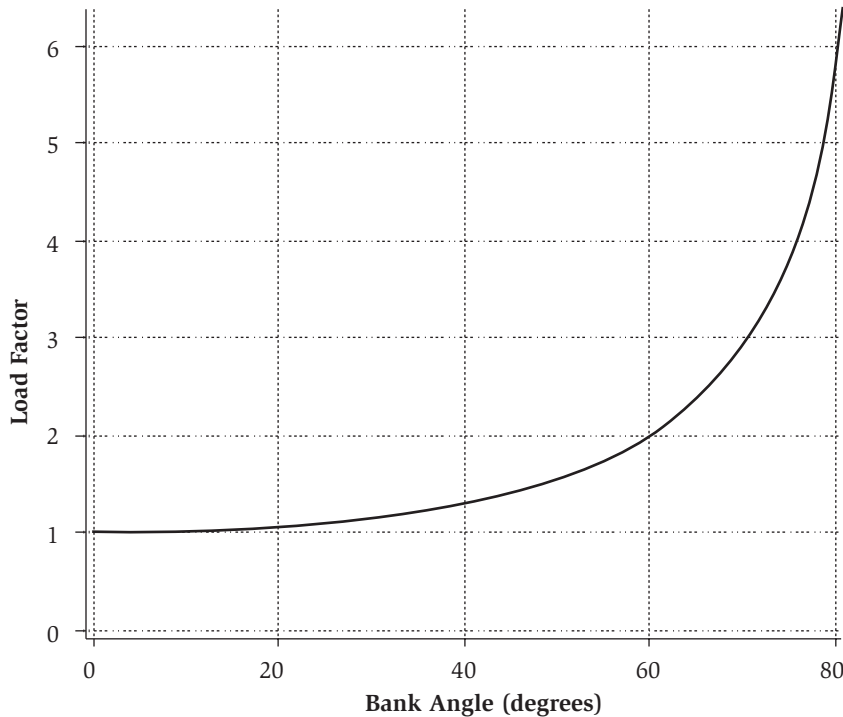


Fig. 1.7. Load factor as a function of bank angle.

load divided by the weight, on the airplane in a 2g turn is 2. Figure 1.7 shows the load factor as a function of bank angle for any airplane in flight. One thing to understand is that the forces on the pilot (or the load) are only related to the *bank angle*, which is the angle made by the wing and the horizon. In Figure 1.6 the bank angle is 60 degrees. The vertical part of the lift must always be equal to the weight of the airplane if the altitude of the airplane does not change during the turn. This is called a *level turn*. So the steeper the bank angle the greater the lift and thus the greater the force felt by the pilot. The insert in the figure shows the forces of a 30 degree turn for comparison. The weight part of the lift is the same, but the other two forces are less. A 2g turn is achieved by banking the airplane at an angle of 60 degrees, independent of the speed of the airplane. Turns will be discussed in more detail in the chapter on “Airplane Performance.”

Straight-and-level flight

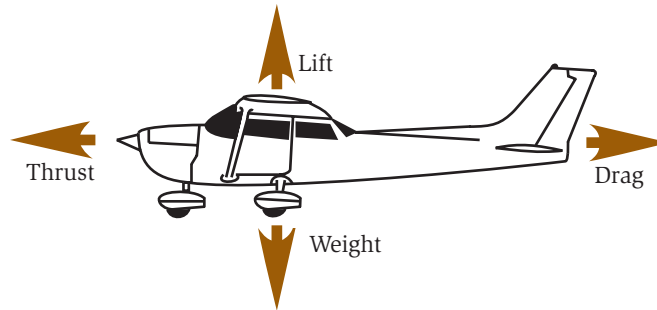


Fig. 1.8. The four forces on an airplane in straight-and-level flight.

The Four Forces

There are four forces associated with the flight of an airplane. These forces, illustrated in Figure 1.8, are lift, weight, thrust, and drag. In *straight-and-level flight* (not changing speed, direction, or altitude) the net lift on the airplane is equal to its weight. We say net lift because, for a conventional airplane design, the horizontal stabilizer pulls down, putting an additional load on the wings. The thrust produced by the engine is equal to the drag, which is caused by air friction and the work done to produce the lift.

The Wright brothers' first flight was less than half the length of a Boeing 777-300.

Mach Number

One important parameter in describing high-speed flight is the *Mach number*. The Mach number is simply the speed of the airplane, or speed of the air, measured in units of the speed of sound. Thus, an airplane traveling at a speed of Mach 2 is going twice the speed of sound. The speed of sound is fundamental for flight because it is the speed of communication between the airplane and the air, and between one part of the air and another. As will be seen in later chapters, as the speed of an airplane approaches Mach 1 there are dramatic changes in its performance. One change that affects performance is that the air

ceases to separate before the wing arrives and instead collides energetically with the wing.

The speed of sound is not a constant in air. In particular it changes with air temperature and thus with altitude. As the air temperature decreases with altitude, so does the speed of sound, though not as quickly. At sea level the value of Mach 1 is about 760 mi/h (1220 km/h). The speed of sound decreases with altitude to about 35,000 ft (about 11,000 m), where the value is about 660 mi/h (1060 km/h). The speed of sound then remains essentially constant to an altitude of 80,000 ft (24,000 m). No airplanes fly above this altitude, with the exception of the Space Shuttle on its way back from space.

The Soviet TU-144 supersonic transport crashed at the Paris Airshow in 1973. It was later found that the pilot was trying to avoid a French Mirage fighter that was trying to take unauthorized pictures of the TU-144 in flight. The TU-144 was planned to compete with the Concorde.

Kinetic Energy

Kinetic energy is the energy of an object because it is moving. The energy difference between a bullet sitting on a table and one flying through the air is the kinetic energy. To be technical, if the bullet had a mass m (say in grams, for example) and were moving at a velocity v , its kinetic energy would be $\frac{1}{2}mv^2$. (There, we have reached the highest level of math complexity that one needs in order to understand flight.)

Because we are going to be discussing the movement of air and the production of propulsion by the acceleration of air or exhaust, it is important that when we say kinetic energy it be understood that we mean the energy due to motion. It is that simple.

Air Pressures

Before we go into understanding flight and airplanes, we should spend just a little time discussing air pressure to remove some common misunderstandings. The following discussion is only strictly true for air moving at speeds below about Mach 0.3 (three-tenths the speed of sound) where the air can be considered incompressible. This will be discussed in greater detail in the next chapter.

Many of us have seen pictures of air passing through a tube that narrows as in Figure 1.9. The figure will often be referred to in text that says something like, “As the area of the tube narrows, the flow velocity must increase. If no other force acts on the fluid, the pressure at point A must be greater than the pressure at point B.” This is the Bernoulli relationship that some are familiar with in the explanation of lift in flight. At first the meaning of “the pressure at point A” seems obvious. What is never said in physics books is that the pressure referred to is measured perpendicular to the direction of flow. It is also not said that there are two other pressures associated with the air at point A. One of them has increased and the other has remained the same. The aeronautical engineers understand this concept, but somehow the information has never made it to the aviation community.

As we have said, there are three pressures associated with flowing air. The first is the *total pressure*. This is measured by bringing the flowing air to a stop. In Figure 1.9, this is measured by placing a tube facing into the airflow. The air stops in the tube and the total pressure

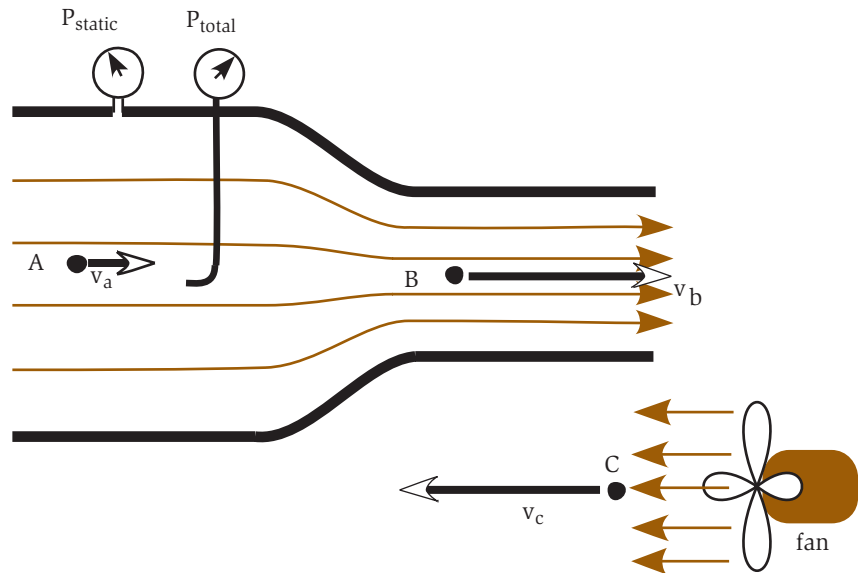


Fig. 1.9. Airflow and pressures.

is measured (P_{total} in the figure). In the situation in the figure, P_{total} is the same at both points A and B. In the language of pilots, this is also known as the *Pitot* pressure, and the figure illustrates a *Pitot tube*.

The second pressure to consider is the *static pressure* (P_{static}), which is measured perpendicular to the airflow through a hole in the wall. This is the pressure most often referred to when the air pressure is discussed in aerodynamics. In the figure the static pressure is higher at point A than at point B.

The third pressure is the *dynamic pressure* (P_{dynamic}), which is the pressure due to the motion of the air, and is a pressure parallel to the flow of air. The dynamic pressure is proportional to the kinetic energy in the air. Thus the faster the air goes the higher the dynamic pressure. This may seem a little complicated, so let us try to put it all together.

The total pressure P_{total} is the sum of the static and dynamic pressures (P_{static} and P_{dynamic}). We have shown how to measure the total, or Pitot, pressure and the static pressure. How is the dynamic pressure of the air measured? Look at the setup in Figure 1.10. Between the tube that measures the total pressure and the tube that measures the static pressure there is placed a *differential pressure gauge*. That is a gauge that measures the difference in pressure between the two ports, which is the difference between total and static pressure. Since static plus dynamic pressure is equal to total pressure, this difference between total and static pressures is the dynamic pressure.

If no energy is added to the air (by some mechanism such as a propeller) the total pressure remains the same and an increase in dynamic pressure causes a decrease in static pressure. So when one reads that the pressure of air decreases because it is going faster, the pressure referred to is the static pressure. But what if energy is added to the air? In the right-hand corner of Figure 1.9 is the picture of a fan. What has happened to the air pressures at point C? The fan is accelerating the air, and thus has done work on the air. Thus the dynamic pressure has increased. Since the air is not confined, the static pressure is the same as in the surrounding environment and thus has

Alexander Graham Bell was a founding member of the Aerial Experiment Association (AEA), which was organized to try to be the first in flight. It was disbanded in 1908 but had been partially successful with the designs of Glenn Curtiss.

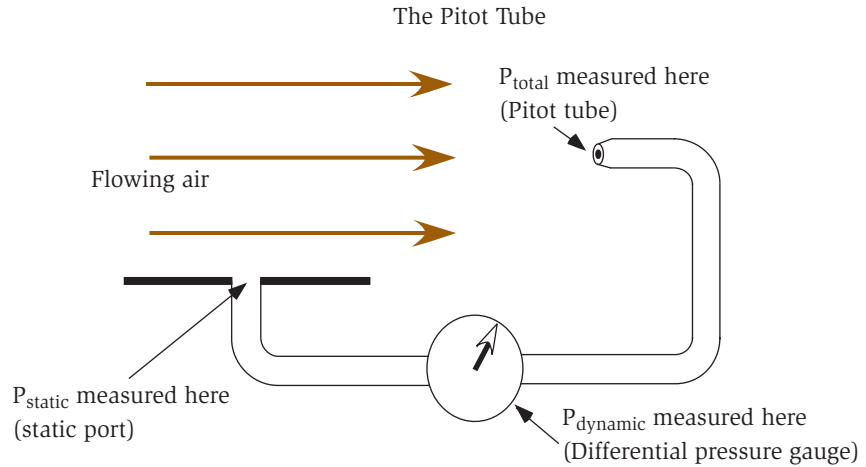


Fig. 1.10. Measurement of pressures with the Pitot tube.

During WWII the length of a belt of 50-caliber machine gun bullets was 27 feet. When a pilot emptied his guns into a single target, he was giving it the “whole nine yards.”

not changed. Thus the total pressure has increased. The moral of this story is that when someone refers to air pressure of moving air they are probably referring to static pressure (though they may not know it). It is also wrong to think that just because air is flowing faster the (static) pressure has decreased. This topic is covered in more detail in the Appendix.

The Pitot Tube

As mentioned previously, the tube measuring the total pressure in Figure 1.10 is called the *Pitot tube*, which, along with static pressure, is used for measuring the airspeed of an airplane. Several Pitot tubes can be seen on the front of the large jets at the airport. A single Pitot tube can be seen protruding (or hanging) from the wing of small airplanes. The hole that measures the static pressure of the air is the *static port*. This port is usually on the side of the airplane fuselage near the front, though it is also occasionally placed on the side of the Pitot tube itself. In an airplane there is a gauge that measures the difference between these two devices and is calibrated in speed. This is the *airspeed indicator*. Since the dynamic pressure measured is related to the dynamic pressure, there is also a correction made by the pilot for the reduced density of the air at altitudes above sea level.

In 1908, a French woman named Therese Peltier became the first woman pilot.

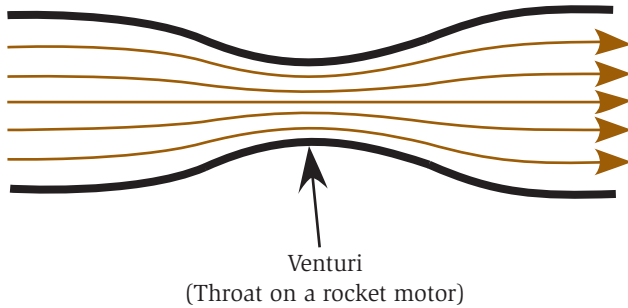


Fig. 1.11. The venturi or throat.

Venturi and Throat

Take a look at Figure 1.11. Here the tube has a constriction in it. In a tube this constriction is often referred to as a *venturi*. If the constriction is in a rocket motor, it is called a *throat*. Off and on in the book we will find occasion to refer to both the venturi and the throat.

Other Books

The purpose of this book is to give the reader a physically intuitive understanding of flight and of aerodynamics, without the use of complicated mathematics. But, at some point one may want to go beyond a merely physical understanding to a level of understanding that does require more mathematics. The authors have found two books very useful in the writing of this book.

The first book is *Aerodynamics for Naval Aviators* by H. H. Hurt, Jr., published by Direction of Commander, Naval Air Systems Command, United States Navy. This book is a wealth of information and insight for those who want to go into greater detail in understanding modern flight.

The second book is *Introduction to Flight*, by John D. Anderson, published by McGraw-Hill. This book is certainly a modern classic in the field of aerodynamics textbooks. *Introduction to Flight* is a serious college textbook and is geared to those with a good understanding of mathematics.

Both are excellent references for those seriously interested in airplanes and aerodynamics. Of the two, *Aerodynamics for Naval*

The world's worst airplane accident was in Tenerife, Canary Islands, on March 27, 1977. Two Boeing 747s collided on the ground, killing 561 people.

Aviators is a better reference for the person interested in a good aeronautical reference without the need of calculus.

Wrapping It Up

The brief discussion of nomenclature, controls, turns, and pressure should give you the necessary tool to understand the rest of the book. In the next chapter, *How Airplanes Fly*, we present a physical description of flight. After flight is understood, we discuss the airplane itself.

How Airplanes Fly

As mentioned in the introduction of this book, a great deal of false concepts and “mythology” have built up around the principles of flight. In this chapter we explain with simple logical discussions the physical phenomena of lift and address some of the errors in the present explanations. Armed with an understanding of lift, we take it further and give you an intuitive understanding of flight in a much broader sense. We start by looking at three descriptions of lift.

The Popular Description of Lift

Most of us have been taught what we will call the “popular description of lift,” which fixates on the shape of a wing. The key point of the popular description of lift is that the air accelerates over the top of the wing. Because of the Bernoulli effect, which relates the speed of the air to the static pressure, a reduced static pressure is produced above the wing, creating lift. The missing piece in the description is an understanding of the cause of the acceleration of the air over the top of the wing. A clever person contributed this piece with the introduction of the “principle of equal transit times,” which states that the air that separates at the leading edge of the wing must rejoin at the

Daniel Bernoulli did not derive the “Bernoulli equation.” His friend Leonid Euler did.

trailing edge. Since the wing has a hump on the top, the air going over the top travels farther. Thus it must go faster to rejoin at the trailing edge. The description is complete.

This is a tidy explanation and it is easy to understand. But one way to judge an explanation is to see how general it is. Here one starts to encounter some troubles. If this description gives us a true understanding of lift, how do airplanes fly inverted? How do symmetric wings (the same shape on the top and the bottom) fly? How does a wing flying at a constant speed adjust for changes in load, such as in a steep turn or as fuel is consumed? One is given more questions than answers by the popular description of lift.

One might also ask if the numbers calculated by the popular description really work. Let us look at an example. Take a Cessna 172, which is a popular, high-winged, four-seat airplane. The wings must lift 2300 lb (1045 kg) at its maximum flying weight. The path length for the air over the top of the wing is only about 1.5 percent greater than the length under the wing. Using the popular description of lift, the wing would develop only about 2 percent of the needed lift at 65 mi/h (104 km/h), which is “slow flight” for this airplane. In fact, the calculations say that the minimum speed for this wing to develop sufficient lift is over 400 mi/h (640 km/h). If one works the problem the other way and asks what the difference in path length would have to be for the popular description to account for lift in slow flight, the answer would be 50 percent. The thickness of the wing would be almost the same as the chord length.

Though enthusiastically taught, there is clearly something seriously wrong with the popular description of lift. The first thing that is wrong is that the principle of equal transit times is not true for a wing with lift. It is true only for a wing without lift. Figure 2.1 shows a computer simulation of the airflow around a wing. Periodically simulated smoke has been introduced to show the changes in the speed of the airflow.

The principle of equal transit times is not true for a wing with lift.

The first thing to notice is that the air going over the top of the wing reaches the trailing edge before the air that goes under the wing. In fact, the air that passed under the wing has a somewhat retarded velocity compared to the velocity of air some distance from the wing. Without the principle of equal

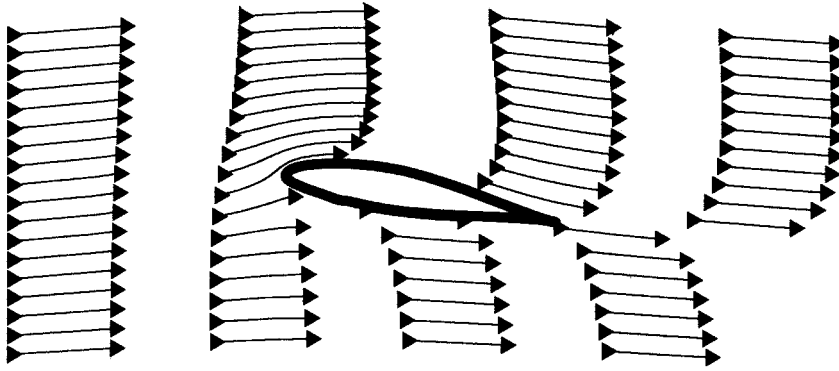


Fig. 2.1. Simulated smoke flowing around an airfoil.

transit times, the popular description of lift loses its explanation for acceleration of the air over the top of the wing.

In explaining the physics of the lift on the wing, we will reveal other problems with this description. But first we would like to introduce two other descriptions of lift. These are the “mathematical description of lift” and the “physical description of lift.” The latter is the description followed in this book.

The Mathematical Description of Lift

Aeronautical engineers use the mathematical description of lift, a name coined for the sake of this discussion. It uses complex mathematics and computer simulations, and is a powerful design tool. Typically, the velocities of the air around the wing are generated with a computer program. Then, using Bernoulli’s equation, the pressures and lift forces are accurately calculated. Since this description often calculates the lift of a wing from the acceleration of the air, this description has quite a bit in common with the popular description of lift.

The aeronautical engineers know that the principle of equal transit times is not true. They often use a mathematical concept called *circulation* to calculate the acceleration of the air over the wing. Circulation is a measure of the apparent rotation of the air around the wing. In the mathematical description of lift circulation is used to

discuss why the air goes more slowly below the wing and faster over the top. While useful for calculations of lift, circulation is not something a pilot needs to consider while flying an airplane. The concept of circulation is useful, however, for the understanding of upwash and of ground effect, and is discussed in greater detail later in this chapter.

Like the popular description of lift, the classical theories in the mathematical description of lift do not illustrate the fact that lift requires the action of a great deal of air, much more than you might imagine. We will not spend much time on the mathematical description of lift, though it will come up from time to time.

The Wright brothers knew nothing of Bernoulli's principle or any mathematical aerodynamic theories. They just watched how birds fly.

The mathematical description of lift is a general term for the analysis tools of classical aerodynamics and computational aerodynamics. If the objective is to accurately compute the aerodynamics of a wing, these are the tools to use, though the description is mathematical and not physical. This is a point lost on many of its proponents. Fortunately, the physical description of lift, presented here, does not require complicated mathematics.

The Physical Description of Lift

The physical description of lift is based primarily on Newton's three laws and a phenomenon called the *Coanda effect*. This description is uniquely useful for understanding the phenomena associated with flight. It is useful for an accurate understanding of the relationships in flight, such as how power increases with load or how the stall speed increases with altitude. It is also a useful tool for making rough estimates ("back-of-the-envelope calculations") of lift. The physical description of lift is also of great use to a pilot who needs an intuitive understanding of how to fly the airplane.

In this description, lift is recognized as a reaction force, that is, wings develop lift by diverting air down. One knows that a propeller produces thrust by blowing air back and that a helicopter develops lift by blowing air down. Propellers and helicopter rotors are simply rotating wings. Thus the concept of a wing diverting air down to produce lift should not be difficult to accept. As we will see, the low

pressure that is formed above the wing accelerates the air down, with almost all of the lift of a well-designed wing coming from the diversion of air from above the wing.

Wings develop lift by diverting air down.

One should be careful not to form the image in their mind of the air striking the bottom of the wing and being deflected down. This is a fairly common misconception, and was also made by Sir Isaac Newton himself. Since Newton was not familiar with the details of airflow over a wing, he thought that the air was diverted down by its impact with the bottom of birds' wings. It is true that there is some lift due to the diversion of air by the bottom of the wing, but the majority of the lift is due to the top of the wing.

The physical description of lift is very powerful. It allows one to have an intuitive understanding of such diverse phenomena as the dependence of lift on the angle of attack of the wing, inverted flight, ground effect, high-speed stalls, and much more. As stated before, this description of lift is that it is useful to the pilot in flight. This cannot be said for the popular description of lift.

Newton's Three Laws

The most powerful tools for understanding flight are Newton's three laws of motion. They are simple to understand and universal in application. They apply to the flight of the lowly mosquito and the motion of the galaxies. Let us look at these laws, not quite in order.

A statement of Newton's first law is:

A body at rest will remain at rest, and a body in motion will continue in straight-line motion unless subjected to an external applied force.

In the context of flight this means that if an object (such as a mass of air), initially motionless, starts to move, there has been force acting on it. Likewise, if a flow of air bends (such as over a wing), there also must be a force acting on it.

Newton's third law can be stated:

For every action there is an equal and opposite reaction.

Eleanor Roosevelt shocked military officers when she told them she wanted to take a ride with an African-American pilot from Tuskegee.

This is fairly straightforward. When you sit in a chair, you put a force on the chair and the chair puts an equal and opposite force on you. Another example is seen in the case of a bending flow of air over a wing. The bending of the air requires a force from Newton's first law. By Newton's third law, the air must be putting an equal and opposite force on whatever is bending it, in this case the wing.

Newton's second law is a little more difficult but also more useful in understanding many phenomena associated with flight. The most common form of the second law is

$$F = ma, \text{ or force equals mass times acceleration}$$

The law in this form gives the force necessary to accelerate an object of a certain mass. For this work we use an alternative form of Newton's second law that can be applied to a jet engine, a rocket, or the lift on a wing. The alternate form of Newton's second law can be stated:

The force (or thrust) of a rocket is equal to the amount of gas expelled per time, times the velocity of that gas.

The amount of gas per time might be in units such as pound mass per second (lbm/s) or kilograms per second (kg/s). The velocity of that gas might be in units such as feet per second (ft/s) or meters per second (m/s). This form of Newton's second law says that if one knows the rate at which gas is expelled from a rocket motor, and the speed of that gas, the thrust of the rocket motor can be easily calculated. To double the thrust, one can double the amount of gas expelled per second, double the velocity of the gas, or a combination of the two.

Let us now look at the airflow around a wing with Newton's laws in mind. In Figure 2.2 we see the airflow around the wing as many of us have been shown at one time or another. Notice that the air approaches the wing, splits, and reforms behind the wing going in the initial direction. This wing has no lift. There is no net *action* on the air and thus there is no lift, the *reaction* on the wing. If the wing has no net effect on the air, the air cannot have any net effect on the wing. Now,

look at another picture of air flowing around a wing (Figure 2.3). First the air comes from below the wing. This is the *upwash*, which will be explained later. The air splits around the wing and leaves the wing at a slight downward angle. This downward-traveling air is the *downwash* and as we will see is the source of lift on a wing. In this figure there has been a net change in the air after passing over the wing. Thus there has been a force acting on the air and a reaction force on the wing. There is lift.

If the wing has no net effect on the air, the air cannot have any net effect on the wing and there is no lift.

The final question to be answered before we can go on to really understand lift is why does the air bend around the wing? The answer is in an interesting phenomenon called the *Coanda effect*.

The Coanda Effect

The Coanda effect has to do with the bending of fluids around an object. For the forces and pressures associated with low-speed flight air is considered not only a fluid but an incompressible fluid. This



Fig. 2.2. Based on Newton's laws, this airfoil has no lift.



Fig. 2.3. The airflow around a real airfoil looks like this.